THE CIRCLE PROPERTY OF $K(\mathbb{Z},1)$, AND THE VANISHING OF $[K(\mathbb{Z},2),K(B,3)]_*$

Purpose and status. Informal proof for the next step of `sag/pi3-s2-postnikov` (handoff `doc/pi3-s2-postnikov-handoff-2.txt`, §3): maps out of $K(\mathbb{Z},1)$ are determined by a point and a loop, and the obligation (H3) of `KZ2Cohomology` follows. Written before the formal proof; each numbered statement is a declaration (§5). It was refereed independently before formalisation: no mathematical error was found; the corrections asked for (directions of retracts, lemmas missing from the library, the step on paths in $\mathsf{L}Y$) are included below. *Status: checked.* `CircleProperty` and `H3Proof` (63 modules, 0 failed); `H3Proof.h3 : KZ2Cohomology.H3`. Every equivalence used below is either given or obtained as a retract; all transfers along an equivalence are proved by induction on it (`Torsor.eqind`), so no half-adjoint coherence is needed. Foundations: those of the development (univalence, images and their surjectivity; function extensionality derived); no higher inductive type and no new axiom. The argument follows the idea of Bezem–Buchholtz–Grayson–Shulman (*Construction of the circle in UniMath*, arXiv:1910.01856): a map out of the torsors is defined by contracting, at every point, a type of equivariant data. Here the torsor structure is used only through the band, so the statement is about an abstract pointed connected type with loop group $\mathbb{Z}$.

1. Setting and conventions

Paths compose in diagrammatic order: for $p : x = y$, $q : y = z$, $p \cdot q : x = z$ is `trans p q`, defined by J on q; hence $p \cdot \mathsf{refl} \equiv p$ and $\mathsf{ap}_F\,\mathsf{refl} \equiv \mathsf{refl}$ hold by definition, while $\mathsf{refl} \cdot p = p$ is a lemma. $\mathbb{Z}$ is `Integers.Int` with suc, $+$ (recursive in the second argument, so $n + 1 \equiv \mathsf{suc}\,n$ by definition).

K is $K(\mathbb{Z},1)$ of the library: $K =$ `EM1[Zab].K1`, the small $\mathbb{Z}$-torsors, pointed by k_0 (the trivial torsor). We use only:

- (K1) K is connected: $\mathsf{isConn}\,2\,K$ (`conn1`); so a family of propositions over K has a section as soon as it holds at k_0 (`Connected.conn-elim`, $n = 0$);
- (K2) the band $\mathsf{band} : (k_0 = k_0) \to \mathbb{Z}$ is an equivalence (`band-equiv`) and a homomorphism, $\mathsf{band}(p \cdot q) = \mathsf{band}\,p + \mathsf{band}\,q$ (`band-hom`).

Let $g : k_0 = k_0$ be the generator, $g = \mathsf{band}^{-1}(1)$, so $\mathsf{band}\,g = 1$. Then, by (K2) and commutativity,

$$(*) \qquad \mathsf{band}(g \cdot p) = 1 + \mathsf{band}\,p = \mathsf{band}\,p + 1 \equiv \mathsf{suc}(\mathsf{band}\,p).$$

For a type Y the free loop type is $\mathsf{L}Y = \Sigma(y : Y).\, y = y$, and

$$E : (K \to Y) \to \mathsf{L}Y, \qquad E\,F = (F\,k_0,\ \mathsf{ap}_F\,g).$$

2. Recursion on $\mathbb{N}$ and on $\mathbb{Z}$

Lemma 1 (recursion on $\mathbb{N}$)**.** *Let W be a type, $\psi : W \to W$ any map and $w : W$. The type*

$$C_0(\psi, w) = \Sigma(u : \mathbb{N} \to W).\, (u\,0 = w) \times \Pi(n : \mathbb{N}).\, u(n+1) = \psi(u\,n)$$

is contractible.

Date: 28 October 2026.

Proof. Centre: $(r, \mathsf{refl}, \lambda n.\mathsf{refl})$ with r defined by recursion, $r\,0 = w$, $r(n+1) = \psi(r\,n)$. Given (u, e_0, e), define $H : \Pi n.\, r\,n = u\,n$ by recursion: $H\,0 = e_0^{-1}$, $H(n+1) = \mathsf{ap}_\psi(H\,n) \cdot e(n)^{-1}$. A path $(r, \mathsf{refl}, \lambda n.\mathsf{refl}) = (u, e_0, e)$ is given by $\mathsf{funext}\,H$ and the two transports, computed by J on a general path $P : r = u$ (then specialised with $\mathsf{happly}(\mathsf{funext}\,H) = H$): the transport of $\mathsf{refl} : r\,0 = w$ is $(\mathsf{happly}\,P\,0)^{-1}$, equal to e_0 for $P = \mathsf{funext}\,H$; the transport of $\lambda n.\mathsf{refl}$ at n is $(\mathsf{happly}\,P\,(n+1))^{-1} \cdot \mathsf{ap}_\psi(\mathsf{happly}\,P\,n)$, which for $P = \mathsf{funext}\,H$ is $(\mathsf{ap}_\psi(H\,n) \cdot e(n)^{-1})^{-1} \cdot \mathsf{ap}_\psi(H\,n) = e(n)$. □

Lemma 2 (recursion on $\mathbb{Z}$)**.** *Let $\varphi : W \to W$ be an equivalence and $w : W$. The type*

$$C_1(\varphi, w) = \Sigma(h : \mathbb{Z} \to W).\, (h\,0 = w) \times \Pi(n : \mathbb{Z}).\, h(\mathsf{suc}\,n) = \varphi(h\,n)$$

is contractible.

Proof. Write $h^+(n) = h(\mathsf{pos}\,n)$, $h^-(m) = h(\mathsf{negsuc}\,m)$. A function on $\mathbb{Z}$, and a Π over $\mathbb{Z}$, is the pair of its restrictions (by cases on $\mathbb{Z}$, and funext). Since $\mathsf{suc}(\mathsf{pos}\,n) \equiv \mathsf{pos}(n+1)$, $\mathsf{suc}(\mathsf{negsuc}\,0) \equiv \mathsf{pos}\,0$ and $\mathsf{suc}(\mathsf{negsuc}(m+1)) \equiv \mathsf{negsuc}\,m$, the condition on h is the conjunction of $h^+(n+1) = \varphi(h^+n)$, $h^+0 = \varphi(h^-0)$ and $h^-m = \varphi(h^-(m+1))$. Hence

$$C_1(\varphi, w) \simeq \Sigma(h^+, e_0, e^+) : C_0(\varphi, w).\, B(h^+0), \qquad B(v) = \Sigma(h^- : \mathbb{N} \to W).\, (v = \varphi(h^-0)) \times \Pi m.\, h^-m = \varphi(h^-(m+1$$

For an equivalence φ with inverse φ^{-1} (its section), $(a = \varphi\,b)$ is a retract of $(b = \varphi^{-1}a)$: by induction on φ it suffices to take $\varphi = \mathrm{id}$, where it is inversion (`equiv-rt`). So $B(v)$ is a retract of $C_0(\varphi^{-1}, \varphi^{-1}v)$, contractible by Lemma 1; and $C_0(\varphi, w)$ is contractible by Lemma 1. A Σ of contractible types over a contractible type is contractible.

As formalised (the "$\simeq$" above taken as retracts in the right direction): C_1 is fibrewise a retract of $\Sigma h.\, \mathrm{Data}'(h)$, the four conditions on h by cases; this is $\Sigma h.\, \Phi(\mathrm{res}\,h)$ for the restriction $\mathrm{res} : (\mathbb{Z} \to W) \simeq (\mathbb{N} \to W)^2$, hence a retract of $\Sigma x.\, \Phi\,x$ (`Retracts.rt-base`, Σ over the domain of the equivalence), which is $\Sigma(a : C_0(\varphi, w)).\, B(a)$ by reassociation. □

3. The circle property

Definition 3. For $(y, l) : \mathsf{L}Y$ and $k : K$ let

$$Q_k = \Sigma(y' : Y).\, \Sigma\big(h : (k_0 = k) \to (y = y')\big).\, \Pi(p : k_0 = k).\; h(g \cdot p) = l \cdot h(p).$$

(The type $k_0 = k$ is the $\mathbb{Z}$-torsor at k, acted on by $g \cdot -$; Q_k is the type of points y' with an equivariant map from it to the paths $y = y'$, $\mathbb{Z}$ acting by $l \cdot -$.)

Lemma 4. *Q_k is contractible for every $k : K$.*

Proof. Contractibility is a proposition, so by (K1) it suffices to treat $k = k_0$. Let $P : ((k_0 = k_0) \to W) \simeq (\mathbb{Z} \to W)$ be precomposition with band^{-1}, whose inverse is $\tilde{h} \mapsto \tilde{h} \circ \mathsf{band}$. A Σ over the domain of an equivalence is a retract of the Σ over its codomain of the family composed with the inverse (`rt-base'`, by induction on P); so $\Sigma h.\, \mathrm{Cond}(h)$ is a retract of $\Sigma \tilde{h}.\, \mathrm{Cond}(\tilde{h} \circ \mathsf{band})$. For $h = \tilde{h} \circ \mathsf{band}$ the condition reads $\Pi p.\, \tilde{h}(\mathsf{band}(g \cdot p)) = l \cdot \tilde{h}(\mathsf{band}\,p)$, a retract (along $(*)$, pointwise) of $\Pi p.\, \tilde{h}(\mathsf{suc}(\mathsf{band}\,p)) = l \cdot \tilde{h}(\mathsf{band}\,p)$, itself a retract of $\Pi n.\, \tilde{h}(\mathsf{suc}\,n) = l \cdot \tilde{h}(n)$ (Π over the equivalence band, `pi-rt`, by induction on it). Hence, adding the singleton $q = \tilde{h}(0)$, Q_{k_0} is a retract of

$$\Sigma(y' : Y).\, \Sigma(q : y = y').\; C_1(l \cdot -,\, q)$$

with $W = (y = y')$ and $\varphi = l \cdot -$, an equivalence. The base $\Sigma(y' : Y).\, y = y'$ is contractible and every $C_1(l \cdot -, q)$ is contractible (Lemma 2). □

Theorem 5 (circle property)**.** *For every type Y, $E : (K \to Y) \to \mathsf{L}Y$ is an equivalence.*

Proof. We show that every fibre $\mathrm{fib}_E(y, l)$ is a retract of $\Pi(k : K).\, Q_k$, which is contractible by Lemma 4 and function extensionality.

(1) Paths in $\mathsf{L}Y$: $\big((a,u)=(b,v)\big)$ is a retract of $\Sigma(e : b = a).\ e \cdot u = v \cdot e$ (`lp-rt`): one way by J on the path in $\mathsf{L}Y$, giving $(\mathsf{refl}, \mathsf{refl} \cdot u = u)$ at refl; the other by based J on e, giving at $e = \mathsf{refl}$ the path $(a,u) = (a,v)$ from $u = \mathsf{refl} \cdot u = v$; the composite is the identity by J and $\mathsf{trans}(p^{-1}, p) = \mathsf{refl}$. So

$$\mathrm{fib}_E(y,l) \ \triangleleft\ \Sigma(F : K \to Y).\, \Sigma(e : y = F\,k_0).\ e \cdot \mathsf{ap}_F\, g = l \cdot e.$$

(2) Distributivity of Π over Σ (an equivalence whose two composites are the identity by definition):

$$\Pi_k\, Q_k \simeq \Sigma(F : K \to Y).\, \Sigma\big(H : \Pi_k\,(k_0 = k) \to (y = F\,k)\big).\, \Pi_k \Pi_p\ H_k(g \cdot p) = l \cdot H_k(p).$$

(3) Yoneda: $j : (y = F\,k_0) \to \Pi_k\,(k_0 = k) \to (y = F\,k)$, $j(e)_k(p) = e \cdot \mathsf{ap}_F\, p$, is an equivalence: its inverse is evaluation at (k_0, refl), with $j(e)_{k_0}(\mathsf{refl}) \equiv e$, and $j(H_{k_0}\mathsf{refl})_k(p) = H_k(p)$ by J on p. A Σ over a base X is a retract of the Σ over a base equivalent to X (`Retracts.rt-base`), so the type of (2) has as a retract

$$\Sigma F.\, \Sigma(e : y = F\,k_0).\, \Pi_k \Pi_p\ e \cdot \mathsf{ap}_F(g \cdot p) = l \cdot (e \cdot \mathsf{ap}_F\, p).$$

(4) Based path induction: for a family D over $\Sigma(k : K).\, k_0 = k$, $D(k_0, \mathsf{refl})$ is a retract of $\Pi_k \Pi_p\, D(k,p)$ (extension by J, evaluation at refl; the composite is the identity by definition). This is the general lemma `yon-rt`, of which `DeloopMaps.Maps.Over.Fixed.yoneda` is an instance. At (k_0, refl) the condition of (3) is, by definition, $e \cdot \mathsf{ap}_F\, g = l \cdot e$.

Composing, $\mathrm{fib}_E(y,l)$ (by (1)) is a retract of the type of (3), hence of $\Pi_k\, Q_k$, which is contractible; so E has contractible fibres (`Whitehead.contr-equiv`). □

Corollary 6 (pointed form)**.** *For a pointed type (Y, y_0) the canonical pointed evaluation $\mathsf{ev}_\circ : \mathrm{Map}_*(K,Y) \to (y_0 = y_0)$, $F \mapsto (\mathsf{loop}\, F)\, g$, is an equivalence, and $\mathsf{ev}_\circ(F, p_F) = p_F^{-1} \cdot \mathsf{ap}_F\, g \cdot p_F$ (`DeloopMaps.Maps.loop-ap`).*

Proof. $\mathrm{Map}_*(K,Y) = \Sigma F.\, F\,k_0 = y_0$ is the fibre over y_0 of $\pi \circ E$, with $\pi : \mathsf{L}Y \to Y$ the first projection, since $\pi(E\,F) \equiv F\,k_0$. As E is an equivalence, $(F,p) \mapsto (E\,F, p)$ is an equivalence onto $\mathrm{fib}_\pi(y_0)$ (by induction on E); and $((y,l),p) \mapsto p_* l$ is an equivalence $\mathrm{fib}_\pi(y_0) \simeq (y_0 = y_0)$, with inverse $l \mapsto ((y_0, l), \mathsf{refl})$ (both round trips by J). The composite sends (F, p_F) to $(p_F)_*(\mathsf{ap}_F\, g)$, and $p_* u = p^{-1} \cdot u \cdot p$ for a loop u (by J on p), which is the formula; so $\mathsf{ev}_\circ$ is homotopic to an equivalence. *Update (§2a of handoff 3):* first formalised as a retract only; now the equivalence (`Circle.pointed-equiv`). A retract of a connected type is connected (`conn-rt`). □

For (H3) only a consequence is used: if $(y_0 = y_0)$ is connected, so is $\mathrm{Map}_*(K,Y)$ (`Deloop.equiv-conn`).

4. (H3)

Theorem 7. *For every abelian group B, $[K(\mathbb{Z},2),K(B,3)]_*$ is contractible (`KZ2Cohomology.H3`).*

Proof. Write $A_2 = K(\mathbb{Z},2)$, $B_3 = K(B,3)$ (`KK[Zab].K two`, `KK[B].K three`), with base points a_0, b_0, and $0 = (\lambda x.\, b_0, \mathsf{refl})$ the constant pointed map. The set of classes is inhabited by [0]; by induction on classes (into the propositions $[f] = [0]$) it suffices that every $f : A_2 \to_* B_3$ is merely equal to 0.

The looping has propositional fibres. For λ = `Maps[A2,a0,B3,b0].loop` : $\mathrm{Map}_*(A_2,B_3) \to \mathrm{Map}_*(\Omega A_2, \Omega B_3)$ the fibre over ϕ is $\mathsf{Unloop}\,\phi = \Sigma F.\, \phi = \lambda F$. By Corollary 10 of `DeloopMaps` (`Levels[c,t].cor10`) with $c = 1$, $t = 1$: A_2 is $\mathsf{isConn}\,3$ (`KK.conn 1`) and B_3 is $\mathsf{isTrunc}\,5 = \mathsf{isTrunc}(2{+}(2{+}1))$ (`KK.trunc 2`); so $\mathsf{Unloop}\,\phi$ is $\mathsf{isTrunc}\,1$, hence a proposition (`HLevels.trunc-one-prop`). (Orientation: $\mathsf{Unloop}\,\phi = \Sigma F.\, \phi = \lambda F$, not fib_λ; only `Unloop` is used.)

The target of the looping is connected. The pointed equivalences $e_1 : \Omega A_2 \simeq_* K(\mathbb{Z},1)$ and $e_2 : \Omega B_3 \simeq_* K(B,2)$ (`KK.e`) give, by pre- and post-composition, $\mathrm{Map}_*(\Omega A_2, \Omega B_3)$ as a retract of $\mathrm{Map}_*(K(\mathbb{Z},1), K(B,2))$ (`H3Proof.rt-ptmap`, by induction on the two pointed equivalences,

`PtEquivOps.ptEqInd`). By Corollary 6, $\mathrm{Map}_*(K(\mathbb{Z},1),K(B,2)) \simeq \Omega K(B,2) \simeq K(B,1)$ (`KK.e 1`), and $K(B,1)$ is connected (`KK.conn 0`). So $\mathrm{Map}_*(\Omega A_2, \Omega B_3)$ is connected (`conn-rt`), and λf, $\lambda 0$ are merely equal.

Conclusion. Given $q : \lambda f = \lambda 0$, the elements (f, refl) and $(0, q)$ of the proposition $\mathtt{Unloop}(\lambda f)$ are equal; their first components give $f = 0$. Since the goal is a proposition, the mere existence of q suffices. □

Remark 8. (1) Only (K1), (K2) are used: the proof applies to any pointed connected type with a homomorphic band onto $\mathbb{Z}$. (2) Theorem 5 is stated for Y in the universe of K; no other size condition appears. (3) For (H4) the same looping has set-level fibres ($t = 2$) and target $\mathrm{Map}_*(K(\mathbb{Z},1),K(B,3)) \simeq \Omega K(B,3) \simeq K(B,2)$; this is not treated here.

5. Correspondence with the formal proof

Files `sag/pi3-s2-postnikov/CircleProperty.seg`, `H3Proof.seg`; `-opens` specification `circle.opens`.

Lemma 1	`NatRec[W,psi,w].contr`
Lemma 2	`IntRec[W,phi,w].contr` (with `equiv-rt`, `rt-data`, `res-equiv`)
$(*)$	`Circle.band-g` (then `g` is locked)
Lemma 4	`Circle[Y].Pt[y,l].Q-contr` (`Q0-rt`, `cond-rt`, `rt-base'`, `pi-rt`)
Theorem 5 (1)–(4)	`Pt.fib-rt` (`lp-rt`), `Pt.X1-rt` (`yon-rt`, `j-equiv`), `Pt.X2-rt`
Theorem 5	`Circle[Y].E-equiv`
Corollary 6	`Circle[Y].evPt`, `evPt-conj`, `pointed-up`, `pointed-equiv`, `ptmap-conn` (`fib-up`, `omega-up`, `subst-conj`, `sub-equiv`)
Theorem 7	`H3Proof.h3 : KZ2Cohomology.H3`